

Day 2: Bren Math Workshop

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Units

UNITS. UNITS. UNITS.

Think about these statements, which all contain the same *value* of 4:

- There are four in the refrigerator.
- There are four **burritos** in the refrigerator.
- There are four **roaches** in the refrigerator.
- There are four **million dollars** in the refrigerator.

UNITS. UNITS. UNITS.

Think about this sign at New Cuyama. Anything odd here?



Units are critical in environmental science

We cannot responsibly work with data without knowing the units of **each variable we're working with.**

That means we need to always **familiarize ourselves with variables**, carefully **check units and any unit conversions**, and understand **how units combine into the units of a dependent variable.**

Metric prefixes

Metric units are built from a **base unit** (like meters, grams, or liters) plus a **prefix** that scales it by a power of 10.

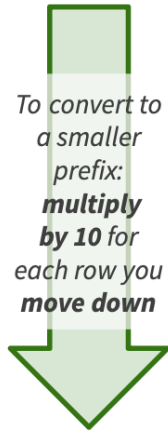
Prefix	Symbol	Factor	Example
kilo	k	10^3	km, kg, kL
hecto	h	10^2	hm, hg, hL
deka	da	10^1	dam, dag, daL
(base unit)	—	10^0	m, g, L
deci	d	10^{-1}	dm, dg, dL
centi	c	10^{-2}	cm, cg, cL
milli	m	10^{-3}	mm, mg, mL

The **factor** indicates how many times to multiply the base unit by 10.

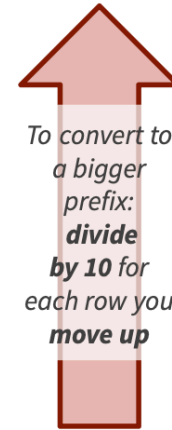
Example: 1 km = 10^3 m = 1000 m, and 1 cm = 10^{-2} m = 0.01 m.

Metric prefixes

Metric units are built from a **base unit** (like meters, grams, or liters) plus a **prefix** that scales it by a power of 10.



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milli	m	10^{-3}	mm, mg, mL



The **factor** indicates how many times to multiply the base unit by 10.

Example: $1 \text{ km} = 10^3 \text{ m} = 1000 \text{ m}$, and $1 \text{ cm} = 10^{-2} \text{ m} = 0.01 \text{ m}$.

Metric prefixes

Example. Convert 750 mL into L.

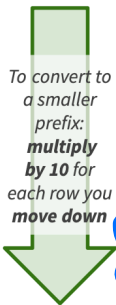
 Let's solve this together.

from mL to L

we divide by $10^3 = 1000$

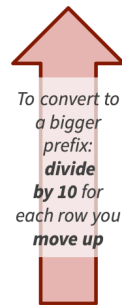
$$750 \text{ mL} = \frac{750}{1000} = 0.75 \text{ L.}$$

To convert to a smaller prefix:
multiply by 10 for each row you **move down**



Prefix	Symbol	Factor	Example
kilo	k	10^3	km, kg, kL
hecto	h	10^2	hm, hg, hL
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Metric prefixes

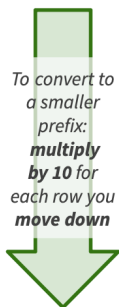
Example. Convert 750 mL into L.

Starting at **milli** and ending at **base**, we move **3 rows up** the table:

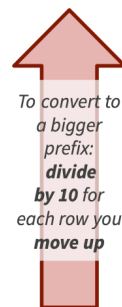
milli → centi → deci → base

So we divide by 10^3 :

$$\begin{aligned} 750 \text{ mL} &= \frac{750}{10^3} \text{ L} \\ &= 0.75 \text{ L} \end{aligned}$$



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Metric prefixes

Exercise. Convert 4.7 km into cm.

 Take a moment to compute it.

multiply by 10^5 :

$$4.7 \text{ km} = 4.7 \cdot 10^5 \text{ cm} \\ = 470,000 \text{ cm}.$$

Prefix	Symbol	Factor	Example
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milli	m	10^{-3}	mm, mg, mL

To convert to a smaller prefix: **multiply by 10** for each row you **move down**

To convert to a bigger prefix: **divide by 10** for each row you **move up**

Metric prefixes

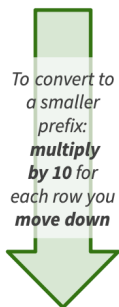
Exercise. Convert 4.7 km into cm.

Starting at **kilo** and ending at **centi**, we move **5 rows down** the table:

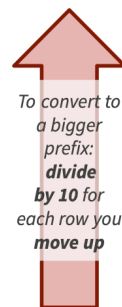
kilo → hecto → deka → base →
deci → centi

So we multiply by 10^5 :

$$\begin{aligned} 4.7 \text{ km} &= 4.7 \times 10^5 \text{ cm} \\ &= 470,000 \text{ cm} \end{aligned}$$



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Converting areas and volumes

So far we've converted **linear** units (length). What about **area** (squared units) and **volume** (cubed units)?

To convert **area** units, **square** the linear conversion factor.

To convert **volume** units, **cube** the linear conversion factor.

Since $1 \text{ m} = 100 \text{ cm}$:

$$* 1 \text{ m}^2 = (100 \text{ cm})^2 = 100^2 \text{ cm}^2 = 10,000 \text{ cm}^2$$

$$1 \text{ m}^3 = (100 \text{ cm})^3 = 100^3 \text{ cm}^3 = 1,000,000 \text{ cm}^3$$

1 m^2 want to convert to cm^2

$$\begin{aligned} 1 \text{ m} &= 100 \text{ cm} \quad \text{then} \quad (1 \text{ m})^2 = (100 \text{ cm})^2 \\ &= 100^2 \text{ cm}^2 = 10^4 \text{ cm}^2 \end{aligned}$$

Converting areas and volumes

Example. Convert 3.5 m^2 into cm^2 .

 Let's solve this together.

Linear conversion is $100 = 10^2$
 $\Rightarrow$ 2 rows down the table.

$$\begin{aligned} 3.5 \text{ m}^2 &= 3.5 \cdot (10^2)^2 \text{ cm}^2 \\ &= 3.5 \cdot 10^4 \text{ cm}^2 \\ &= 35,000 \text{ cm}^2 \end{aligned}$$

To convert to a smaller prefix: multiply by 10 for each row you move down

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To convert to a bigger prefix: divide by 10 for each row you move up

$$(x^a)^b = x^{ab}$$

$$x^a \cdot x^b = x^{a+b}$$

Converting areas and volumes

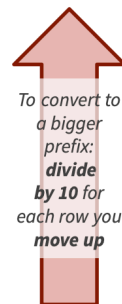
Example. Convert 3.5 m^2 into cm^2 .

The linear factor from meters to centimeters is 10^2 (2 rows down the metric prefix table). For area, we **square** this factor:

$$\begin{aligned} 3.5 \text{ m}^2 &= 3.5 \times (10^2)^2 \text{ cm}^2 \\ &= 3.5 \times 10^4 \text{ cm}^2 \\ &= 35,000 \text{ cm}^2. \end{aligned}$$



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Converting areas and volumes

Exercise. Convert 2 m^3 into cm^3

 Take a moment to solve this.

linear conversion: 10^2

$$2 \text{ m}^3 = 2 \cdot (10^2)^3 \text{ cm}^3$$

$$= 2 \cdot 10^{2 \cdot 3} \text{ cm}^3$$

$$= 2 \cdot 10^6 \text{ cm}^3$$

$$= 2,000,000 \text{ cm}^3$$

To convert to a smaller prefix: multiply by 10 for each row you move down

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Converting areas and volumes

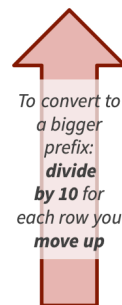
Exercise. Convert 2 m^3 into cm^3 .

The linear factor from meters to centimeters is 10^2 (2 rows down the metric prefix table). For volume, we **cube** this factor:

$$\begin{aligned} 2 \text{ m}^3 &= 2 \times (10^2)^3 \text{ cm}^3 \\ &= 2 \times 10^6 \text{ cm}^3 \\ &= 2,000,000 \text{ cm}^3. \end{aligned}$$



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Dimensional analysis (unit conversions)

In dimensional analysis, we multiply initial units by a sequence of **unit conversion factors** to arrive at the final desired units.

Example: How many centimeters are there in 6 inches?

 Let's solve this together.

We know $1 \text{ inch} = 2.54 \text{ cm}$. Given us two conversion factors:

$\frac{2.54 \text{ cm}}{1 \text{ inch}}$

and

$\frac{1 \text{ inch}}{2.54 \text{ cm}}$

* important: numerator and denominator represent same quantity.
then multiply by the appropriate conversion factor:

$$6 \text{ in} = 6 \text{ in} \cdot \frac{2.54 \text{ cm}}{1 \text{ in}} = 6 \cdot \frac{2.54}{1} \cancel{\text{in}} \cdot \frac{\text{cm}}{\cancel{\text{in}}}$$

$$= 6 \cdot 2.54 \text{ cm} = 15.24 \text{ cm}.$$

Dimensional analysis (unit conversions)

In dimensional analysis, we multiply initial units by a sequence of **unit conversion factors** to arrive at the final desired units.

Example: How many centimeters are there in 6 inches?

We know: 1 inch = 2.54 cm. This gives us two unit conversion factors:

$$\frac{2.54 \text{ cm}}{1 \text{ in}} \text{ and } \frac{1 \text{ in}}{2.54 \text{ cm}}.$$

Important: the fraction the numerator and denominator represent the same quantity.

Then we multiply by the *appropriate* conversion factor and simplify:

$$\begin{aligned}
 6 \text{ in} &= 6 \text{ in} \cdot \frac{2.54 \text{ cm}}{1 \text{ in}} \\
 &= 6 \frac{2.54}{1} \cdot \text{in} \frac{\text{cm}}{\text{in}} \\
 &= 15.24 \cancel{\text{in}} \frac{\text{cm}}{\cancel{\text{in}}} \\
 &= 15.24 \text{ cm}
 \end{aligned}$$

Dimensional analysis (unit conversions)

Example. Convert $100 \frac{\text{g}}{\text{cm}^3}$ into units of $\frac{\text{kg}}{\text{in}^3}$, given that $1 \text{ cm}^3 = 0.061 \text{ in}^3$.

 Let's solve this together.

Conversion factor for g : $\frac{1 \text{ kg}}{1000 \text{ g}}$

Conversion factor for cm^3 : $\frac{1 \text{ cm}^3}{0.061 \text{ in}^3}$

$$\begin{aligned} 100 \frac{\text{g}}{\text{cm}^3} &= 100 \frac{\text{g}}{\text{cm}^3} \cdot \frac{1 \text{ cm}^3}{0.061 \text{ in}^3} \cdot \frac{1 \text{ kg}}{10^3 \text{ g}} \\ &= \left(100 \cdot \frac{1}{0.061} \cdot \frac{1}{1000} \right) \cdot \left(\frac{\cancel{\text{g}}}{\cancel{\text{cm}^3}} \cdot \frac{\text{cm}^3}{\text{in}^3} \cdot \frac{\text{kg}}{\cancel{\text{g}}} \right) \\ &= 1.639 \frac{\text{kg}}{\text{in}^3} . \end{aligned}$$

Dimensional analysis (unit conversions)

Example. Convert $100 \frac{\text{g}}{\text{cm}^3}$ into units of $\frac{\text{kg}}{\text{in}^3}$, given that $1 \text{ cm}^3 = 0.061 \text{ in}^3$.

We need one unit conversion factor for **g**: $\frac{1 \text{ kg}}{1000 \text{ g}}$.

And another one for **cm³**: $\frac{1 \text{ cm}^3}{0.061 \text{ in}^3}$.

Then we multiply by the conversion factors and simplify

$$\begin{aligned} 100 \frac{\text{g}}{\text{cm}^3} &= 100 \frac{\text{g}}{\text{cm}^3} \cdot \frac{1 \text{ cm}^3}{0.061 \text{ in}^3} \cdot \frac{1 \text{ kg}}{1000 \text{ g}} \\ &= 100 \frac{1}{0.061} \frac{1}{1000} \cdot \frac{\text{g}}{\text{cm}^3} \frac{\text{cm}^3}{\text{in}^3} \frac{\text{kg}}{\text{g}} \\ &= \frac{1}{0.61} \cdot \frac{\cancel{\text{g}}}{\cancel{\text{cm}^3}} \frac{\cancel{\text{cm}^3}}{\text{in}^3} \frac{\text{kg}}{\cancel{\text{g}}} \\ &= 1.639 \frac{\text{kg}}{\text{in}^3}. \end{aligned}$$

Parts per million (ppm)

Parts per million (ppm) is another way of writing a fraction (just like percentage) but suited to very small concentrations (e.g. a contaminant in water or soil).

 $1\% = \frac{1}{100}$, and $1\text{ ppm} = \frac{1}{1,000,000}$

We need 10,000 ppm to get 1%

so $1\% = 10,000 \text{ ppm}$.

* To convert percent $\rightarrow$ ppm: multiply percentage by 10,000 = 10^4

To convert ppm $\rightarrow$ percent: divide ppm by 10,000.

Parts per million (ppm)

Parts per million (ppm) is another way of writing a fraction (just like percentage) but suited to very small concentrations (e.g. a contaminant in water or soil).

We have that:

$$1\% = \frac{1}{100} \qquad 1 \text{ ppm} = \frac{1}{1,000,000}$$

From these fractions we see we need 10,000 ppm to get 1%, so **1% = 10,000 ppm**.

To convert **percent** → **ppm**, multiply percentage by 10,000.

To convert **ppm** → **percent**, divide ppm by 10,000.

Parts per million (ppm)

Example. Convert 0.008% to ppm.

 Let's solve this together.

converting percent $\rightarrow$ ppm : multiply by 10,000.

$$\begin{aligned} 0.008\% &= 0.008 \times 10,000 \text{ ppm} \\ &= \underline{80 \text{ ppm.}} \end{aligned}$$

Parts per million (ppm)

Example. Convert 0.008% to ppm.

We are converting percent → ppm, so we multiply by 10,000:

$$\begin{aligned} 0.008\% &= 0.008 \times 10,000 \text{ ppm} \\ &= 80 \text{ ppm} \end{aligned}$$

Unit conversion practice

1. Convert 120 ppm to a percentage.
2. Convert 3,400 mm to meters.
3. Convert an area of 3.2 km^2 to m^2 .
4. A stock solution is 4 g/L. Express this in mg/mL.
5. Convert $8.1 \frac{\text{km}}{\text{s}}$ to miles per hour, given that 1 km = 0.621 miles.
6. A lake has average depth 800 cm and surface area 2.5 km^2 . Using Volume = depth $\times$ area, find the volume in m^3 .

Unit conversion practice

 Let's see a solution!

1. Convert 120 ppm to a percentage.

$$\cancel{120} \quad 120 \text{ ppm} = 120 / 10^4 = 0.012\%$$

2. Convert 3,400 mm to meters. $3400 \text{ mm} = \frac{3400}{1000} = 3.4 \text{ m}$

3. Convert an area of 3.2 km² to m². $1 \text{ km} = 10^3 \text{ m}$
 $1 \text{ km}^2 = 10^6 \text{ m}^2$
 $3.2 \text{ km}^2 = 3.2 \cdot 10^6 \text{ m}^2$

Unit conversion practice

4. A stock solution is 4 g/L. Express this in mg/mL.

 Let's see a solution!

$$\frac{4 \cancel{\text{g}}}{1 \cancel{\text{L}}} \times \frac{1 \cancel{\text{K}}}{1000 \cancel{\text{mL}}} \times \frac{1 \cancel{\text{Kg}}}{1000 \cancel{\text{g}}} = 4 \frac{\text{mg}}{\text{mL}}$$

Unit conversion practice

$$\frac{\cancel{s}}{\cancel{s}} = 1$$

5. Convert $8.1 \frac{\text{km}}{\text{s}}$ to miles per hour, given that $1 \text{ km} = 0.621 \text{ miles}$.

 Let's see a solution!

$$\begin{aligned} & \frac{8.1 \cancel{\text{km}}}{\cancel{s}} \cdot \frac{60 \cancel{s}}{1 \text{ min}} \cdot \frac{60 \text{ min}}{1 \text{ hr}} \cdot \frac{0.621 \text{ mi}}{1 \cancel{\text{km}}} \\ & = 18,108.36 \frac{\text{mi}}{\text{hr}} \end{aligned}$$

Unit conversion practice

6. A lake has average depth 800 cm and surface area 2.5 km^2 . Using Volume = depth $\times$ area, find the volume in m^3 .

 Let's see a solution!

$$800 \cancel{\text{cm}} \times \frac{\text{m}}{100 \cancel{\text{cm}}} \times 2.5 \cancel{\text{km}^2} \times \frac{10^6}{\cancel{\text{km}^2}}$$

$$800 \times 2.5 \times 10^6 \times \text{m}^3$$

$2 \times 10^7 \text{ m}^3$

Unit conversion practice: solutions

1. $120 \text{ ppm} = 120/10,000 = 0.012\%$

2. $3,400 \text{ mm} = 3,400/10^3 = 3.4 \text{ m}$

3. $3.2 \text{ km}^2 \times (1000 \text{ m/km})^2 = 3.2 \times 10^6 \text{ m}^2$

4. $4 \text{ g/L} = 4000 \text{ mg}/1000 \text{ mL} = 4 \text{ mg/mL}$

5. $8.1 \frac{\text{km}}{\text{s}} \times 3600 \frac{\text{s}}{\text{hr}} \times 0.621 \frac{\text{mi}}{\text{km}} = 18108.36 \frac{\text{mi}}{\text{hr}}$

6. $\text{Area} = 2.5 \times 10^6 \text{ m}^2$. $\text{Volume} = 8 \times 2.5 \times 10^6 = 2 \times 10^7 \text{ m}^3$

Let's take a 10 minute break



image: Flaticon.com

Functions

Functions

Functions are mathematical expressions that tell us how input values are related to output values.

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For example, $y = 3x - 5$ is a function that tells us the **value of y** at **any value of x** . In this scenario, we would probably say **y is a function of x** .

Functions

Functions are mathematical expressions that tell us how input values are related to output values.

For example, $y = 3x - 5$ is a function that tells us the **value of y** at **any value of x** . In this scenario, we would probably say **y is a function of x** .

Could you also rewrite it and say **x is a function of y** ?

Here, with no knowledge of what's an input and what's an output, sure - but usually in environmental science we specify the input variable(s), and the output variable(s) carefully.

What follows is the expression in the format of:

“[output variable(s)] is/are a function of [input variable(s)]”.

Also: output variable = **dependent variable** and input variable = **independent variable**.

Thinking about inputs and outputs

For the following combinations of related variables, which do you expect would be the **input** and the **output** in a function describing how they are related? E.g.

“Evapotranspiration is a function of air temperature.”

 Write your answer individually as **a sentence**.

1. fuel (biomass) / slope / wildfire severity / windspeed / air temperature
2. wind speed / power generated by wind turbine

Function notation



single variable function:

$$f(x) = [\text{expression containing only } x]$$

x the only variable

multivariate function:

$$g(\underline{a}, \underline{T}, \underline{z}) = [\text{expression containing } a, T \text{ and } z]$$

Function notation

Single variable (univariate) function:

$$f(x) = [\textit{expression containing } x]$$

Multivariate function:

$$g(a, T, z) = [\textit{expression containing } a, T, \text{ and } z]$$

Evaluating functions

We evaluate functions by plugging in values of the input variables.

 Example:

Evaluate $g(x, t) = 2.4x + 0.5t^2$ at $x = 3$ and $t = 10$

$$\begin{aligned} g(3, 10) &= 2.4 \cdot 3 + 0.5(10)^2 \\ &= \underline{\underline{57.2}} \end{aligned}$$

Evaluating functions

We evaluate functions by plugging in values of the input variables.



Example:

Evaluate $g(x, t) = 2.4x + 0.5t^2$ at $x = 3$ and $t = 10$

$$g(3, 10) = 2.4(3) + 0.5(10^2) = 57.2$$

Graphs

Why graphs?

Graphs are a way for us to more easily process trends or patterns that may be more challenging to understand in a table or list.

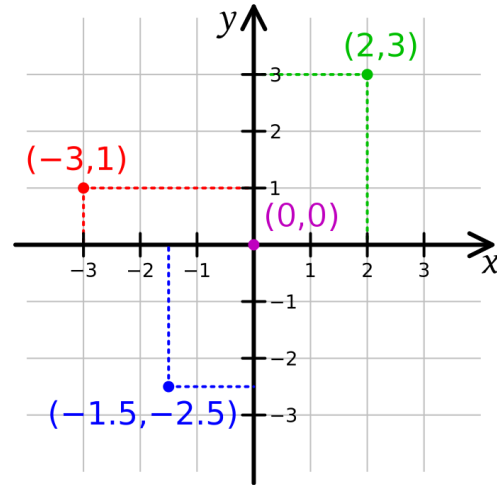
Which could be easier to understand?

The xy coordinate system

Graphs generally move in a 2-dimensional plane with a coordinate system using two axes:

- x -axis (horizontal axis)
- y -axis (vertical axis)

Axes units must be defined depending on the application.



The xy coordinate system

We can join series of points to make graphs

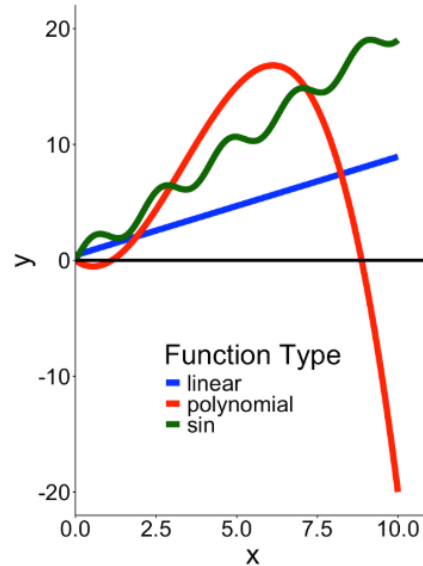
Functions on a single variable and a single output can be graphed

Two key ingredients to graphs

Intercepts

- x -intercept
 - Where does the graph intersect the x -axis?
 - In other words, for what values of x do we have $(x, 0)$ be part of the graph?

What are the x and y intercepts of the polynomial function in red?

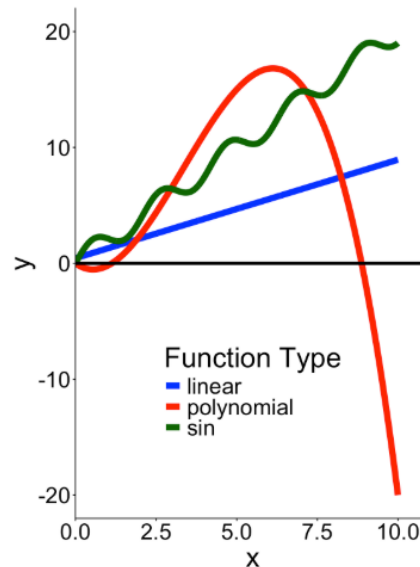


Two key ingredients to graphs

Intercepts

- x -intercept
 - Where does the graph intersect the x -axis?
 - In other words, for what values of x do we have $(x, 0)$ be part of the graph?
- y -intercept
 - Where does the graph intersect the y -axis?
 - In other words, for what value of y do we have $(0, y)$ be part of the graph?

What are the x and y intercepts of the polynomial function in red?

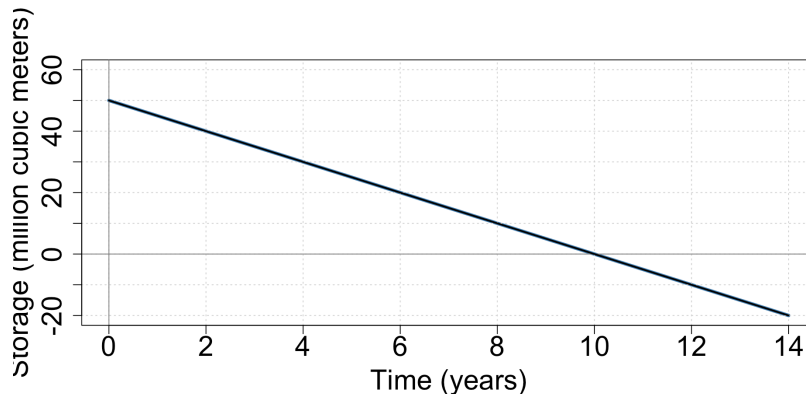


Intercepts practice

 Solve these individually.

1. A reservoir is being drained for irrigation. The graph shows its storage volume (in millions of cubic meters) as a function of time (in years).

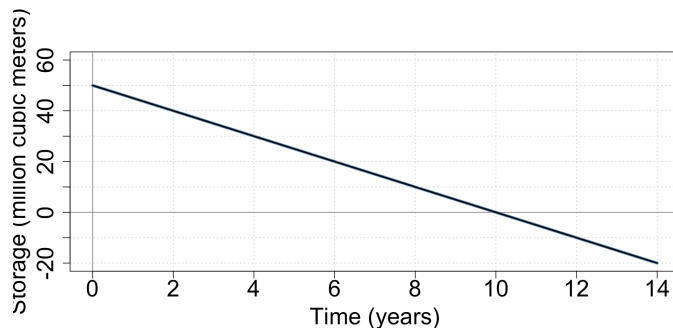
- When will the reservoir be empty? Are you using the x or y intercept?
- What is the initial volume of the reservoir? Are you using the x or y intercept?



Intercepts practice (1)

- When will the reservoir be empty? Are you using the x or y intercept?
- What is the initial volume of the reservoir? Are you using the x or y intercept?

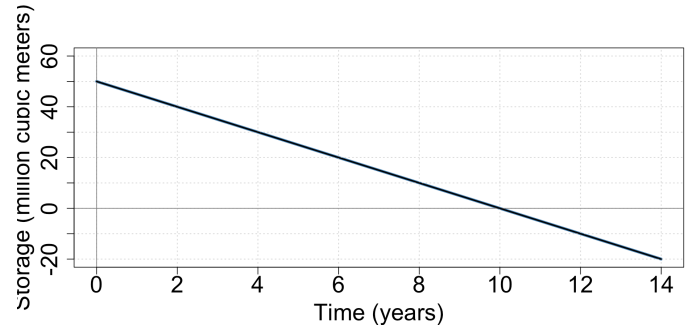
 Let's see a solution!



Intercepts practice (1)

- When will the reservoir be empty? Are you using the x or y intercept?
- What is the initial volume of the reservoir? Are you using the x or y intercept?

 Let's see a solution!



The reservoir will be empty after 10 years. To identify this we look at the x -intercept, which happens at $(10, 0)$.

The initial volume of water in the reservoir is 50 cubic meters. To identify this we look at the y -intercept, which happens at $(0, 50)$.

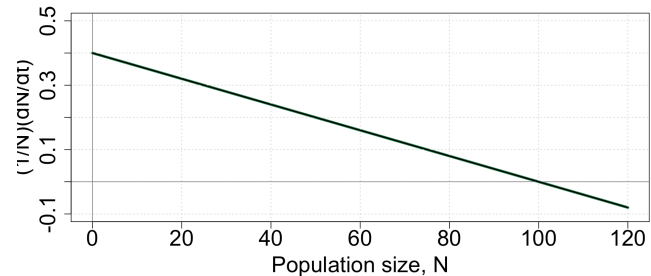
Intercepts practice (2)

Population size is denoted by N , this is the total number of individuals.

Per-capita growth rate, $\frac{1}{N} \frac{dN}{dt}$, is a function of N .

- What does the y -intercept of $\frac{1}{N} \frac{dN}{dt}$ represent?
- What does its x -intercept represent?

 Let's see a solution!



Intercepts practice (2)

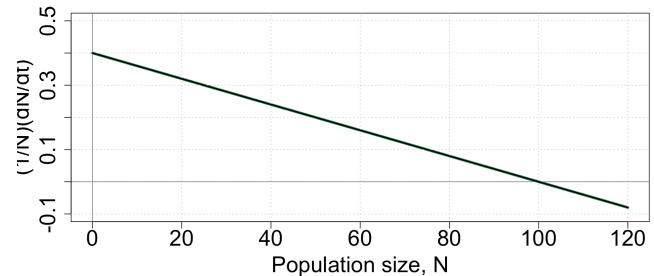
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- What does its x -intercept represent?

 Let's see a solution!

- y -intercept = the population's maximum per-capita growth rate when it is small and not yet resource-limited.
- x -intercept = the population size at which per-capita growth stops.



“Reading” graphs

Understanding graphs

When you look at graphs, the first things you should ask:

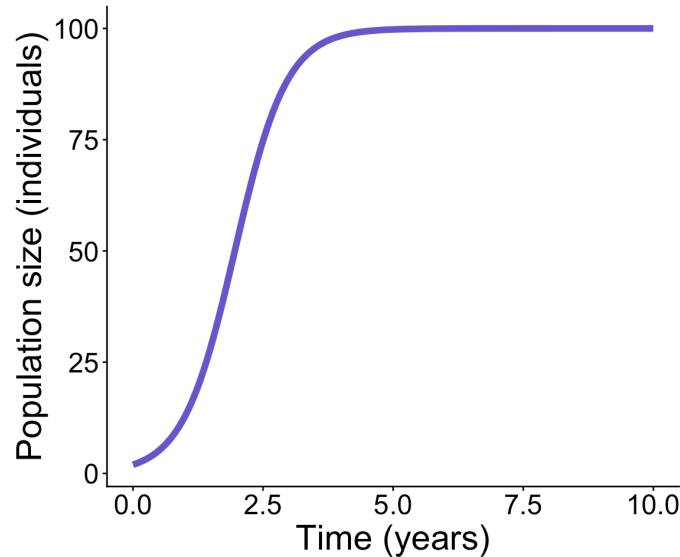
- What variables are being plotted (e.g. x- & y-axis, including units)?
- What values are plotted (e.g. raw values, transformed, means, etc.)?
- What are the overall takeaways (e.g. trends, maxima, minima, axes intercepts) and am I understanding them responsibly?

Reading graphs (in general)

“On the x-axis we have [x-variable] measured in [units] and on the y-axis the the [y-variable] measured in [units]. This figure shows the [change/pattern/relationship] between [x-variable] and [y-variable]. Overall [overall statement of pattern / trend / findings].”

Reading graphs (in general)

“On the x-axis we have [x-variable] measured in [units] and on the y-axis the the [y-variable] measured in [units]. This figure shows the [change/pattern/relationship] between [x-variable] and [y-variable]. Overall [overall statement of pattern / trend / findings].”





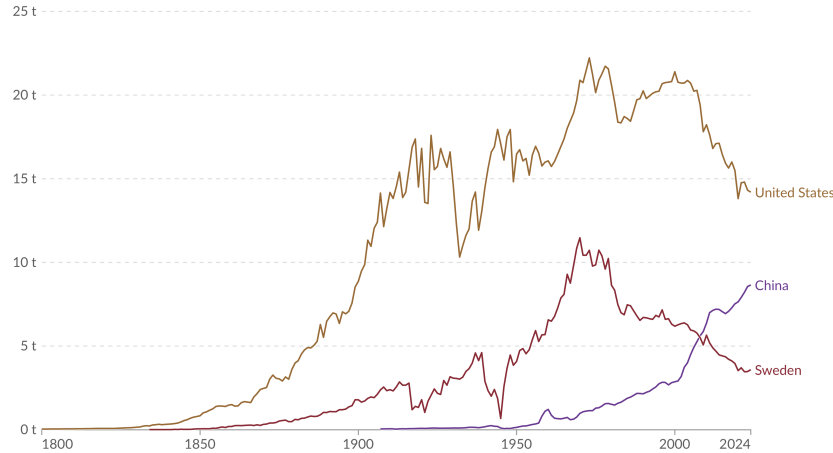
Partner 1: read the graph to your partner.

“On the x-axis we have [x-variable] measured in [units] and on the y-axis the the [y-variable] measured in [units]. This figure shows the [change/pattern/relationship] between [x-variable] and [y-variable]. Overall [overall statement of pattern / trend / findings].”

CO₂ emissions per capita

Carbon dioxide (CO₂) emissions from burning fossil fuels and industrial processes. This includes emissions from transport, electricity generation, and heating, but not land-use change.

Our World
in Data



Data source: Global Carbon Budget (2025); Population based on various sources (2024) CO₂ measured in tons per person.
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

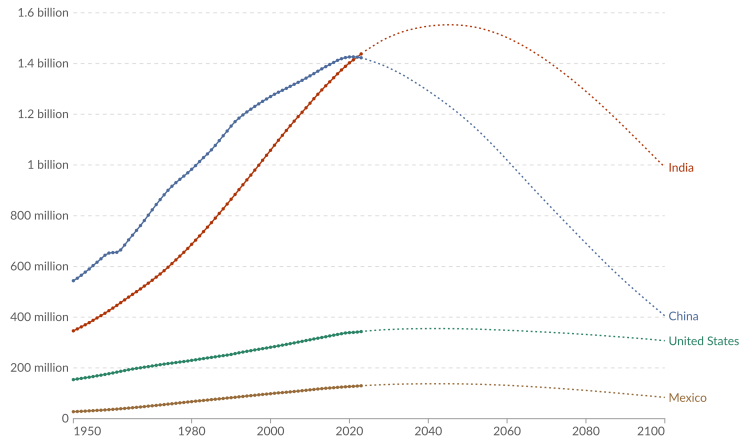


Partner 2: read the graph to your partner.

“On the x-axis we have [x-variable] measured in [units] and on the y-axis the the [y-variable] measured in [units]. This figure shows the [change/pattern/relationship] between [x-variable] and [y-variable]. Overall [overall statement of pattern / trend / findings].”

Population

Future projections are based on the UN's low scenario¹.



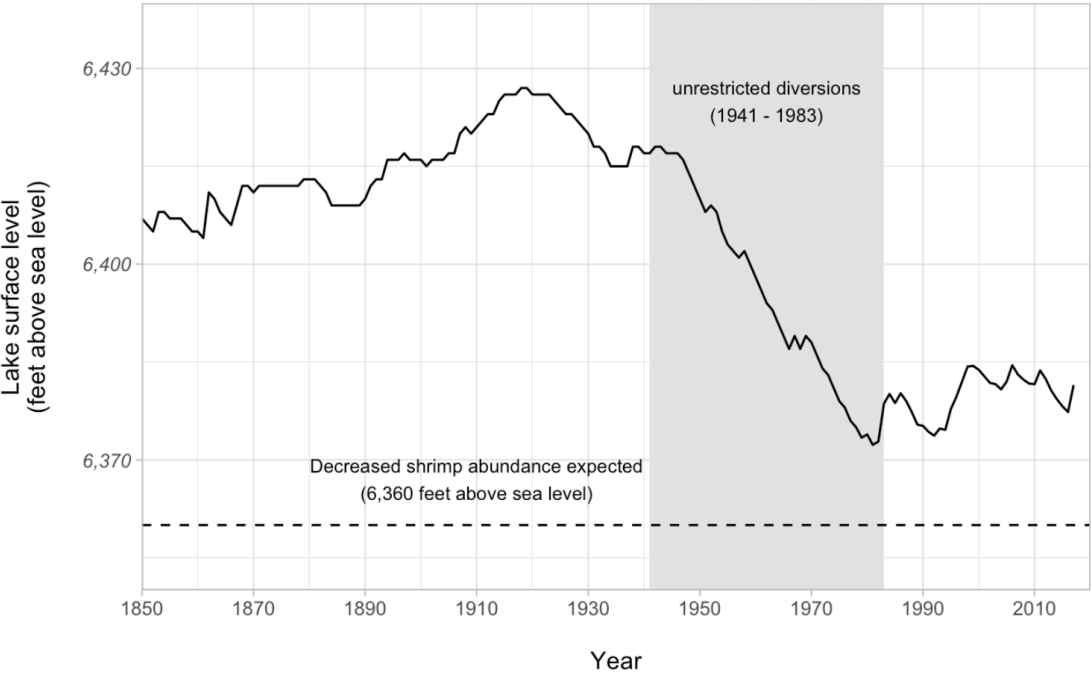
Data source: UN, World Population Prospects (2024)

OurWorldinData.org/population-growth | CC BY

1. **United Nations projection scenarios** The United Nations' World Population Prospects include a range of projected scenarios for population change. These scenarios rely on different assumptions for fertility, mortality, and migration patterns to project different demographic futures.
[Read more on population.un.org](https://population.un.org/wpp/)

Reading graphs practice

Mono Lake levels (1850 - 2017)



Data: Mono Basin Clearinghouse

Wrapping up...

Get some extra practice!

Units & graphs problem set



What we covered today

-  **Metric prefixes**
-  **Area & volume conversions**
-  **Dimensional analysis (unit conversions)**
-  **Parts per million (ppm)**
-  **Definition of functions**
-  **Graphing**
-  **x and y intercepts**
-  **Reading graphs**
-  **Applications to MESM courses**